

Urjasvi Suthar

+91 97374-35861 | urjasvisuthar@gmail.com | linkedin.com/in/urjasvi-suthar | [Github](#) | [Website](#) | Gujarat, India

EXPERIENCE

USDKit

Sept. 2024 – May 2025

Graphics Engineer Intern

Remote

- Codebase consisted of C, C++, Objective-C, Metal/Vulkan and Pixar's USD 3D file format.
- **Developed text rendering system** using multiple advanced algorithms, improving visual quality and performance across different hardware configurations.
- **Researched font legibility** improvements for enhancing user readability across different screen sizes and DPI.
- **Built multiple advanced animation systems** (Caches, Transforms, Skeletal Animations), with variable FPS and refresh rate. Validated results against industry tools like Blender and Maya. **All in < 1ms**.
- **Debugged complex graphics issues** using professional development tools such as XCode/RenderDoc, GDB/LLDB and custom visualization tools, reducing bug resolution time by 40%.

RealityBob

June 2023 – Feb. 2024

Software Engineer Intern

Remote

- Worked with C++, JS, WebAssembly and WebGPU.
- **Led porting process of native graphics application to web platform**, enabling browser-based 3D graphics.
- **Bridged native and web technologies** by writing custom integration code between **multi-threaded C++** graphics engine and modern web standards, solving compatibility challenges.
- **Documented porting process** and usage of various techniques encountered along.
- **Worked independently with minimal supervision** along the porting process.
- **Evaluated long-term feasibility of web-based 3D graphics** for business applications through research and prototyping, informing strategic product decisions.

PROJECTS

ZigCPURasterizer

website/articles/zigcpurasterizer

- **Designed entire rasterization pipeline** with multi meshes, lighting, materials loaded from glTF format.
- **Implemented physically-based rendering (PBR) pipeline** with LTC area lights and shadow mapping.
- **Achieved real-time performance with 14.27x speedup** through algorithmic improvements (AABB, backface culling) and data structure redesign.
- **Built memory efficient mesh data-structure** for cache-friendly vertex processing, **reducing time by 1.14x**.
- **Documented entire optimization journey** with detailed performance analysis and profiling results.

EDUCATION

Indian Institute Of Information Technology (IIIT), Sri City

AP, India

Bachelor of Technology (B.Tech)

Aug. 2021 – Aug. 2025

EXTRACURRICULAR ACTIVITIES

Club Enigma

Aug. 2023 – Jul. 2024

Computer Graphics Lead

IIIT-S

- **Taught computer graphics fundamentals to 7+ junior students**, covering rendering concepts and practical implementation techniques.
- **Mentored student projects across multiple domains**, providing technical guidance and hands-on collaboration to help them deliver working prototypes.
- **Conducted various sessions**, industry guest talks and major events like **Global Game Jam** under the club.

CLASSES

Courses: Computer Architecture (CA), Computer Graphics & Multimedia (CGM), Virtual/Augmented Reality (VAR), Compiler Design (CD), High Performance Computing (HPC).

SKILLS & INTERESTS

Programming: C, C++, Objective-C, Zig, Python, JavaScript, C#

Technologies: Metal, OpenGL, WebGPU, GLSL, AArch64, LLDB/GDB, XCode, RenderDoc, Instruments, MacOS, UNIX, WASM, Git, Github, Linux, CMake